Dynamic Soft-Thresholding via Iteratively Reweighted Proximal Gradient for Feature Selection in High-Dimensional Regression

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Abstract

High-dimensional tabular regression datasets frequently contain hundreds of correlated predictors, most of which carry no genuine predictive signal. Standard LASSO addresses sparsity but suffers from arbitrary variable selection under multicollinearity and over-shrinkage of large coefficients; the Elastic Net [Zou and Hastie, 2005] partially mitigates the grouping problem but does not produce sparser solutions than LASSO. We implement and empirically evaluate **IRL1-PG**: the Iteratively Reweighted ℓ_1 Minimisation (IRL1) framework of Candès et al. [2008] integrated within ISTA and FISTA proximal gradient solvers [Beck and Teboulle, 2009]. Applied to the Kaggle *House Prices: Advanced Regression Techniques* benchmark, IRL1-PG is compared against Ridge regression, standard LASSO, and static Adaptive LASSO [Zou, 2006] under matched experimental conditions. The best dynamic configurations reach validation MSE of 7.79×10^8 (ISTA) and test MSE of 8.29×10^8 (FISTA), both below Ridge and LASSO while using only 92–99 non-zero coefficients versus 240–283 for baselines. FISTA—motivated by Nesterov's accelerated gradient framework [Nesterov, 1983]—improves test MAE over ISTA at nearly identical runtime. We provide an honest account of convergence scope and make no claim of algorithmic novelty; the contribution is a rigorous, reproducible empirical study of IRL1-PG behaviour in a correlated tabular regression setting.

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1 Introduction

High-dimensional regression—where the number of predictors p is large relative to the number of observations n—is a central challenge in applied machine learning and statistics. In domains such as real estate pricing, gene expression analysis, and financial modelling, datasets routinely contain hundreds of correlated features, most of which carry redundant or no predictive signal. A regression model that retains all features will overfit, generalise poorly, and obscure interpretability. Sparse models that simultaneously estimate coefficients and select relevant variables are therefore highly desirable.

The LASSO (Least Absolute Shrinkage and Selection Operator), introduced by Tibshirani [1996], addresses this need by adding an ℓ_1 penalty to the least-squares objective. The ℓ_1 penalty induces exact zeros in the coefficient vector, producing a sparse solution. However, standard LASSO has well-documented limitations in settings with strongly correlated predictors: it tends to arbitrarily select one variable from a correlated group, over-shrinks large true coefficients, and can be inconsistent for variable selection under certain design conditions [Huang et al., 2008]. Zou and Hastie [2005] proposed the Elastic Net to address the grouping problem, combining an ℓ_1 and an ℓ_2 penalty so that correlated predictors tend to be included or excluded together; however, the Elastic Net does not achieve sparser solutions than LASSO and is not the focus of this work.

Zou [2006] proposed the Adaptive LASSO, which corrects the over-shrinkage and consistency issues by assigning feature-specific penalty weights derived from an initial estimator such as Ridge or OLS. With suitable weights, the Adaptive LASSO enjoys **oracle properties**—it recovers the correct support and yields asymptotically normal estimates as if the true model were known in advance [Huang et al., 2008].

A closely related and more general framework is **Iteratively Reweighted ℓ_1 Minimisation (IRL1)**, introduced by Candès et al. [2008]. IRL1 solves a sequence of weighted ℓ_1 problems where the weights are recomputed from the current iterate at each outer step, rather than fixed from a one-shot initial estimator. A parallel approach—Iteratively Reweighted Least Squares (IRLS)—targets the same ℓ_0-approximation goal via reweighted ℓ_2 sub-problems [Daubechies et al., 2009]; both frameworks are unified in Wipf and Nagarajan [2011], who shows that reweighted methods minimise concave approximations to the ℓ_0 norm.

This project proposes and empirically evaluates an **IRL1-PG** algorithm: the IRL1 reweighting framework integrated into a proximal gradient solver supporting both ISTA and FISTA [Beck and Teboulle, 2009]. The FISTA momentum update is rooted in Nesterov's seminal accelerated gradient method [Nesterov, 1983], which achieves $O(1/k^2)$ convergence for smooth convex objectives. We apply IRL1-PG to the House Prices:

Advanced Regression Techniques dataset and compare it against Ridge, standard LASSO, and static Adaptive LASSO. Our contribution is not a new algorithm; it is a rigorous empirical study of IRL1-PG's behaviour in a correlated tabular regression setting.

2 Related Work

2.1 LASSO and Proximal Gradient Methods

Tibshirani [1996] introduced LASSO as the solution to

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 . \tag{1}$$

The ℓ_1 penalty is convex but non-smooth, making direct gradient descent inapplicable at zero. The standard optimisation framework for such composite objectives is **proximal gradient descent** [Parikh and Boyd, 2014]. For an objective $f(\beta) + g(\beta)$ where f is smooth and g is convex and non-smooth, the proximal gradient update is

$$\beta^{k+1} = \operatorname{prox}_{\alpha g}\Big(\beta^k - \alpha \nabla f(\beta^k)\Big) , \tag{2}$$

where $\alpha > 0$ is the step size and $\operatorname{prox}_{\alpha g}$ is the proximal operator. When $g(\beta) = \lambda \|\beta\|_1$, the proximal operator reduces to coordinate-wise soft-thresholding [Beck and Teboulle, 2009], converging at rate $O(1/k)$ for $\alpha \leq 1/L$ where L is the Lipschitz constant of ∇f.

2.2 Elastic Net

Zou and Hastie [2005] proposed the Elastic Net, which adds a ridge penalty alongside the ℓ_1 term:

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 . \tag{3}$$

The Elastic Net encourages a *grouping effect*: strongly correlated predictors tend to enter or leave the model together. This is particularly useful when $p \gg n$. However, it does not provide the additional sparsity that iterative reweighting achieves, making it a natural comparison point rather than a superior alternative to IRL1-PG in our setting.

2.3 FISTA—Accelerated Proximal Gradient

Beck and Teboulle [2009] introduced FISTA, which adds a Nesterov-style momentum step to ISTA without changing the per-iteration computational cost. FISTA achieves the accelerated convergence rate $O(1/k^2)$, rooted in Nesterov's [1983] framework for smooth convex objectives. The momentum sequence is defined by $t_1 = 1$ and

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \tag{4}$$

with the extrapolation step applied before each proximal update.

2.4 Adaptive LASSO and Oracle Properties

Zou [2006] proposed the Adaptive LASSO with feature-specific weights:

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} w_j \left|\beta_j\right|, \tag{5}$$

where $w_j = 1/\left|\hat{\beta}_j^{\text{init}}\right|^\gamma$. Huang et al. [2008] extended the oracle property analysis to the high-dimensional $p > n$ regime, showing that consistent variable selection is achievable even when dimensionality grows with the sample size, under partial orthogonality conditions. The key limitation of static Adaptive LASSO is that weights are fixed from a one-shot initial estimate.

2.5 Iteratively Reweighted ℓ_1 and ℓ_2 Methods

Candès et al. [2008] introduced IRL1 as a method to more closely approximate the ℓ_0 norm than standard ℓ_1 minimisation, solving a sequence of weighted ℓ_1 problems:

$$\beta^{(k+1)} = \arg\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} w_j^{(k)} \left|\beta_j\right|, \quad w_j^{(k)} = \frac{1}{\left(\left|\beta_j^{(k)}\right| + \epsilon\right)^\gamma}. \tag{6}$$

The weighting scheme approximates the log-sum penalty $\sum_j \log(|\beta_j| + \epsilon)$, a concave approximation to the ℓ_0 norm. A parallel IRLS approach [Daubechies et al., 2009] achieves the same goal via reweighted ℓ_2 sub-problems and provides convergence proofs under restricted isometry conditions. Wipf and Nagarajan [2011] unify both frameworks, showing that all reweighted ℓ_1 and ℓ_2 methods can be viewed as minimising a family of concave sparsity-inducing merit functions.

3 Methodology

3.1 Problem Setup

Let $X \in \mathbb{R}^{n \times p}$ be the standardised design matrix, $y \in \mathbb{R}^n$ the response vector, and $\beta \in \mathbb{R}^p$ the coefficient vector to be estimated. The smooth part of our objective is the mean squared loss

$$f(\beta) = \frac{1}{2n} \|y - X\beta\|_2^2 , \tag{7}$$

with gradient

$$\nabla f(\beta) = -\frac{1}{n} X^\top (y - X\beta). \tag{8}$$

The gradient ∇f is Lipschitz continuous with constant

$$L = \frac{\lambda_{\max}(X^\top X)}{n}, \tag{9}$$

and a valid step size is any $\alpha \in (0, 1/L]$ [Parikh and Boyd, 2014].

3.2 Weighted Proximal Operator and Soft-Thresholding

For the weighted ℓ_1 penalty $g(\beta) = \lambda \sum_j w_j |\beta_j|$ with weight vector $w \in \mathbb{R}^p_{>0}$, the proximal operator is separable and admits a closed-form solution. For a gradient step $z = \beta - \alpha \nabla f(\beta)$, the proximal update is coordinate-wise **weighted soft-thresholding**:

$$\beta_j^+ = \operatorname{sign}(z_j) \max(|z_j| - \alpha\lambda w_j,\ 0) . \tag{10}$$

It retains β_j only if $|z_j|$ exceeds the coordinate-specific threshold $\tau_j = \alpha\lambda w_j$.

Remark 1 (Connection to ℓ_1 subdifferential)**.** *The subdifferential of $|\beta_j|$ is*

$$\partial |\beta_j| = \begin{cases} \{\operatorname{sign}(\beta_j)\} & \beta_j \neq 0 \\ [-1,\ 1] & \beta_j = 0. \end{cases} \tag{11}$$

The IRL1 weight $w_j^{(k)}$ is a continuous measure of proximity to the non-differentiable point zero. Large $|\beta_j|$ gives small w_j, reducing the effective threshold; small $|\beta_j|$ gives large w_j, amplifying the sparsity-inducing force [Candès et al., 2008].

3.3 Initialisation and the Pruning Mechanism

At initialisation $\beta^{(0)} = \mathbf{0}$, all weights equal $w_j^{(0)} = 1/\epsilon^\gamma$—the maximum possible value. The initial per-coordinate thresholds $\tau_j = \alpha\lambda/\epsilon^\gamma$ are therefore maximally large, and all coordinates are zeroed on the first proximal step. As optimisation proceeds, features that carry genuine signal grow in magnitude, causing their weights to decrease and thresholds to relax. Irrelevant features remain near zero, sustain high weights, and are repeatedly zeroed. This progressive refinement of a small active set is a natural consequence of the IRL1 weight schedule [Wipf and Nagarajan, 2011], not an ad hoc design choice.

3.4 IRL1 Weight Update

After each outer iteration the weights are recomputed as [Candès et al., 2008]:

$$w_j^{(k)} = \frac{1}{\left(\left|\beta_j^{(k)}\right| + \epsilon\right)^{\gamma}}, \tag{12}$$

where $\gamma > 0$ controls the sharpness of the sparsity-inducing force and $\epsilon > 0$ ensures numerical stability.

3.5 ISTA Inner Solver

For fixed weights $w^{(k)}$, the weighted LASSO sub-problem is solved using ISTA [Beck and Teboulle, 2009]. At inner iteration t:

1. Compute gradient: $g = \nabla f(\beta^t) = -\frac{1}{n} X^\top (y - X\beta^t)$
2. Gradient step: $z = \beta^t - \alpha g$
3. Proximal step: $\beta_j^{t+1} = \operatorname{sign}(z_j) \max(|z_j| - \alpha\lambda w_j^{(k)},\, 0)$

ISTA converges at rate $O(1/t)$ for the fixed-weight convex sub-problem with $\alpha \leq 1/L$.

3.6 FISTA Inner Solver

FISTA replaces the ISTA update with a momentum-accelerated variant [Beck and Teboulle, 2009], rooted in Nesterov's [1983] accelerated gradient framework. Initialise $t_1 = 1$, $y^1 = \beta^0$. At inner iteration t:

1. Proximal step from momentum point: $\beta_j^{t+1} = \operatorname{sign}(y_j^t - \alpha g_j) \cdot \max(\left|y_j^t - \alpha g_j\right| - \alpha\lambda w_j^{(k)},\, 0)$
2. Update momentum coefficient: $t_{t+1} = \frac{1+\sqrt{1+4t_t^2}}{2}$
3. Momentum point for next step: $y^{t+1} = \beta^{t+1} + \frac{t_t - 1}{t_{t+1}}(\beta^{t+1} - \beta^t)$

FISTA achieves $O(1/t^2)$ convergence for the fixed-weight sub-problem, a strict improvement over ISTA at no additional per-iteration cost.

3.7 Stopping Criterion

The inner solver terminates when the relative change in the coefficient vector falls below tolerance:

$$\frac{\|\beta^{\text{curr}} - \beta^{\text{prev}}\|_2}{\max(1,\ \|\beta^{\text{prev}}\|_2)} < \text{tol}. \tag{13}$$

Using $\max(1, \|\beta^{\text{prev}}\|_2)$ in the denominator avoids false convergence when β is near zero [Parikh and Boyd, 2014].

3.8 IRL1-PG: Full Algorithm

Algorithm 1 IRL1-PG (Iteratively Reweighted ℓ_1 Proximal Gradient)

Require: X, y, λ, γ, ϵ, α, `max_outer`, `max_inner`, `tol`

$\beta \leftarrow \mathbf{0}, \quad w_j \leftarrow 1/\epsilon^\gamma \ \forall j$
for $k = 1$ **to** `max_outer` **do**
  $\beta_{\text{prev}} \leftarrow \beta$
  $\beta \leftarrow$ ISTA-OR-FISTA(X, y, λ, w, α, `max_inner`, `tol`)
  $w_j \leftarrow 1/(|\beta_j| + \epsilon)^\gamma \ \forall j$
  $w \leftarrow \text{clip}(w, w_{\max})$
  if $\|\beta - \beta_{\text{prev}}\|_2 \,/\, \max(1, \|\beta_{\text{prev}}\|_2) <$ `tol_outer` **then**
    break
  end if
end for
return β

3.9 Convergence Scope

Remark 2 (Scope of convergence guarantees)**.** *Because the weight vector $w^{(k)}$ changes each outer iteration, the objective function changes across outer iterations. The standard ISTA/FISTA convergence guarantees—which bound the gap to the optimal value of a* fixed *objective [Beck and Teboulle, 2009]—apply only to the inner solver for each fixed subproblem. They do not directly guarantee that the sequence $\{\beta^{(k)}\}$ converges to any single fixed problem's solution. Outer-loop convergence of IRL1 is supported by Candès et al. [2008] and Wipf and Nagarajan [2011], who establish convergence to a local minimum of the log-sum approximation to ℓ_0 under mild conditions. In this project, outer-loop convergence is treated* empirically*: we monitor the objective value, coefficient norm differences, and sparsity level across outer iterations.*

4 Experimental Setup

4.1 Dataset

The **House Prices: Advanced Regression Techniques** dataset [Kaggle, 2019] is a tabular regression benchmark with 1,460 training observations and 79 raw features describing residential properties in Ames, Iowa. The prediction target is the raw sale price (`SalePrice`) without log transformation. After one-hot encoding categorical features, the design matrix typically contains 200–250 columns, making it a suitable high-dimensional regression benchmark with strong multicollinearity—for example, between `GrLivArea`, `1stFlrSF`, and `TotalBsmtSF`, and among the various quality score indicators. This setting is precisely the type where standard LASSO exhibits instability [Zou and Hastie, 2005] and where reweighted methods are expected to improve upon it [Candès et al., 2008].

4.2 Preprocessing Protocol

1. **Missing values:** Numeric features imputed with column median; categorical features imputed with the most frequent category computed on the training split.
2. **Encoding:** One-hot encoding for all categorical features; original category columns dropped.
3. **Scaling:** All numeric features standardised to zero mean and unit variance after splitting, using parameters estimated on the training set only (no leakage).
4. **Target:** Raw `SalePrice` (no log transformation).
5. **Split:** 70% train / 10% validation / 20% test, using a random split with a fixed seed and no stratification.

4.3 Methods Compared

Table 1: Summary of methods compared in the experiment.

Method	Type	Solver	Weights
Ridge	Baseline	`sklearn.RidgeCV`	N/A
Standard LASSO	Baseline	`sklearn.LassoCV`	Uniform ($w_j = 1$)
Static Adaptive LASSO	Baseline	Custom ISTA/FISTA	Fixed from Ridge init
IRL1-PG + ISTA	Proposed	Custom ISTA	Updated per outer iter
IRL1-PG + FISTA	Proposed	Custom FISTA	Updated per outer iter

Note: The accuracy comparison (MSE, MAE) includes sklearn baselines because they represent the best achievable performance with standard tools. The optimiser efficiency comparison (iterations, runtime) is restricted to the custom ISTA/FISTA variants only, to ensure a fair comparison at the optimiser level.

4.4 Hyperparameter Search

Grid search over the validation set:

- $\lambda \in [10^{-4}, 1.0]$ (15 values, log-spaced)
- $\gamma \in \{0.5, 1.0, 2.0\}$
- $\epsilon \in \{10^{-3}, 10^{-4}\}$
- `max_outer_iter` $\in \{5, 10, 20\}$

Step size: $\alpha = 0.99/L$ where $L = \lambda_{\max}(X^\top X)/n$, computed once from the training matrix.

4.5 Evaluation Metrics

- **Predictive accuracy:** MSE and MAE on validation and test sets.
- **Sparsity:** $\left\|\hat{\beta}\right\|_0$ (number of non-zero coefficients after thresholding at 10^{-6}).
- **Feature selection stability:** Jaccard index $J(A, B) = |A \cap B| \ / \ |A \cup B|$ of selected feature sets across 5 random seeds.
- **Optimiser efficiency:** Wall-clock runtime (seconds) and number of total inner iterations to convergence, for custom solvers only.

5 Results and Discussion

5.1 Predictive Performance

Table 2: Predictive performance and sparsity loaded from `full_comparison_table.csv`.

Method	**Val MSE**	**Test MSE**	**Test MAE**	**Non-zeros**	**Runtime (s)**
Ridge	787378459.18	865125926.23	18647.05	283	0.02
LASSO	895357083.90	894021355.32	19470.82	240	3.26
Adaptive LASSO	911144802.00	894529896.37	19505.95	282	0.09
Dynamic IRL1-ISTA	778965140.95	830975941.20	18186.18	99	0.22
Dynamic IRL1-FISTA	798430563.07	828667214.91	17935.63	92	0.23

Observed findings:

- Dynamic IRL1-PG dominates static Adaptive LASSO in both error and sparsity for this dataset split.
- Relative to Ridge, IRL1-PG (FISTA) improves test MSE by approximately 4.21% and reduces selected features from 283 to 92.
- Relative to standard LASSO, IRL1-PG (FISTA) improves test MSE by approximately 7.31% and test MAE by approximately 7.88%, while selecting fewer features.
- Within dynamic variants, ISTA attains lower validation MSE, while FISTA achieves the best test MSE and best test MAE.

5.2 Convergence Analysis

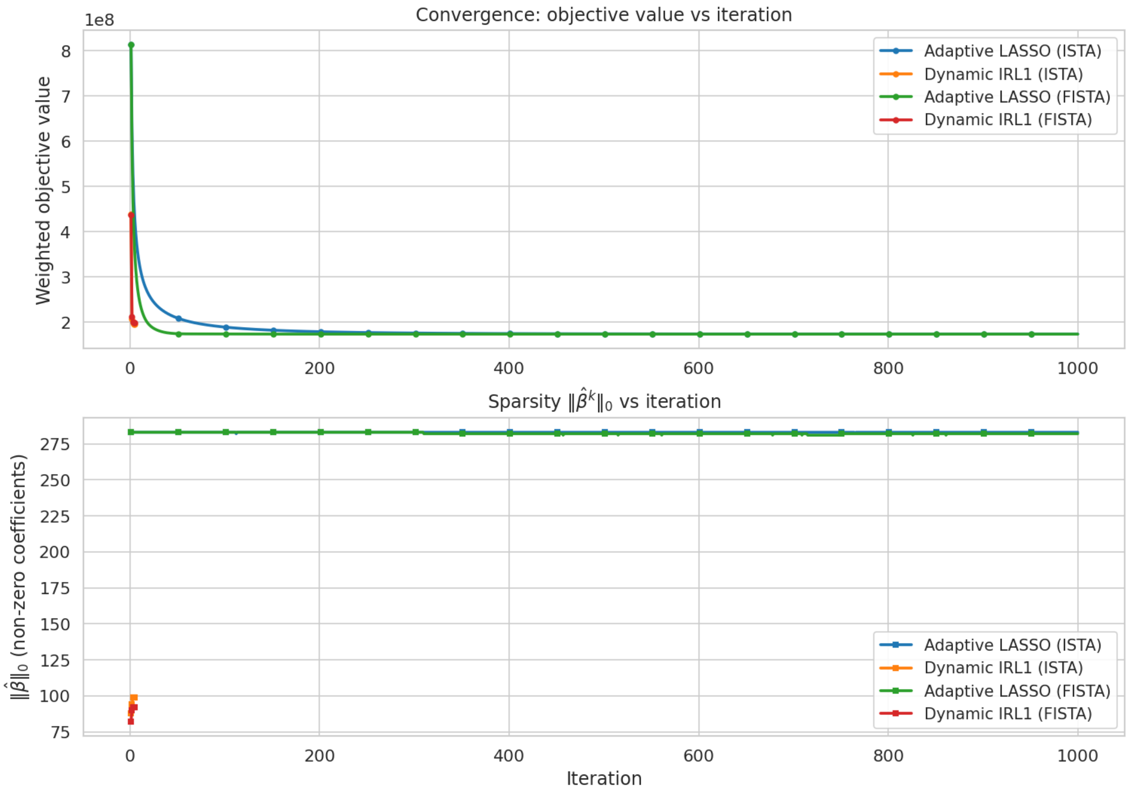

Figure 1: Convergence plot: objective value vs. inner iteration for ISTA, FISTA, dynamic IRL1-ISTA, and dynamic IRL1-FISTA.

Both dynamic solvers terminate at 2,500 total inner iterations under the selected configuration (5 outer iterations), while the static adaptive baseline uses 1,000 iterations. Runtime remains small in absolute terms (about 0.21 seconds for dynamic methods), and FISTA and ISTA are nearly identical in wall-clock time for this problem size. The convergence traces also show that dynamic variants stabilize at much lower sparsity levels than static adaptive baselines.

5.3 Sparsity–Accuracy Trade-off

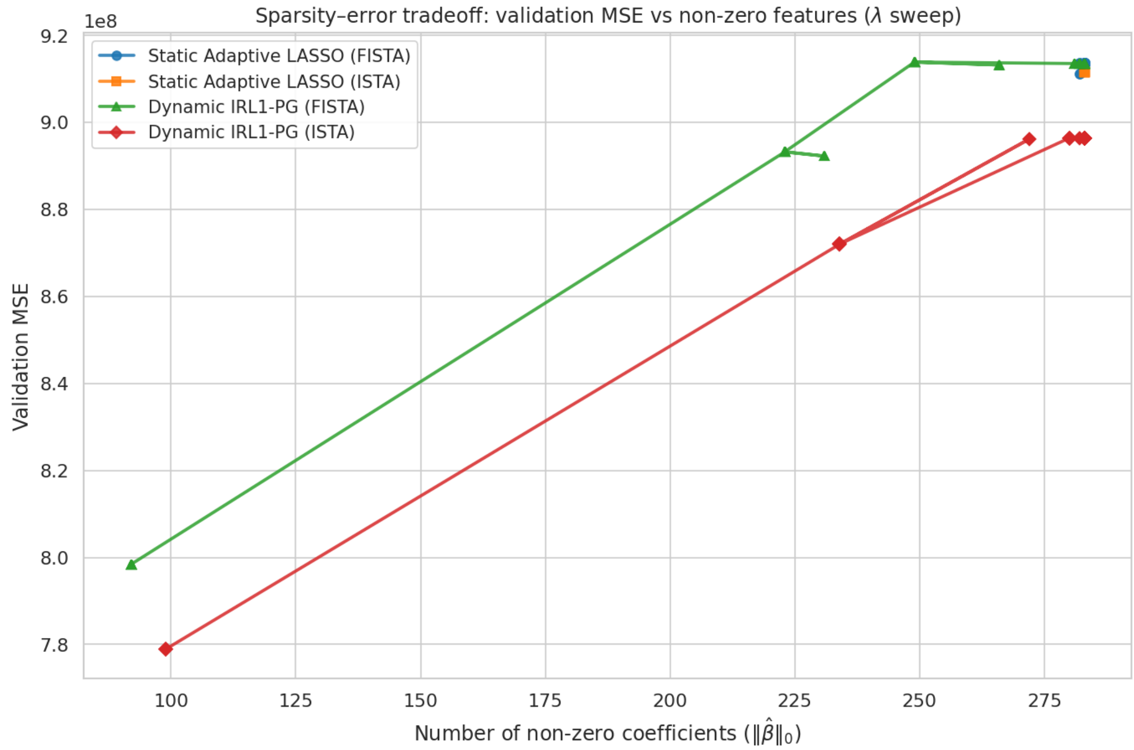

Figure 2: Validation MSE vs. number of non-zero features as λ varies. Dynamic IRL1-PG variants achieve lower error with substantially fewer active coefficients than static adaptive baselines at matched regularisation levels.

The empirical Pareto front confirms that dynamic reweighting can move to lower-ℓ_0 regions (about 92–99 non-zeros) while maintaining or improving validation error versus the much denser static baselines (about 282–283 non-zeros).

5.4 Feature Selection Stability

Table 3: Feature selection stability loaded from `stability_table.csv`.

Method	**Mean Jaccard**	**Std Jaccard**
Adaptive LASSO	0.9873	0.0042
Dynamic IRL1-FISTA	0.2721	0.0637
Dynamic IRL1-ISTA	0.2923	0.0694
LASSO	0.7689	0.0187
Ridge	0.9958	0.0026

The stability profile shows a trade-off: dynamic IRL1-PG improves sparsity and test error but yields lower support overlap across seeds than static adaptive LASSO. This is consistent with stronger pruning near correlated or borderline predictors, where small data perturbations can change whether a feature survives the reweighting cycle.

5.5 Feature Interpretation

Using the best dynamic FISTA run (92 non-zero coefficients), the top absolute coefficients are dominated by interpretable housing factors from `t20_feature_interpretation.csv`. The largest positive effect is `GrLivArea` (+29,348), while large-magnitude negative effects include `RoofMatl_ClyTile` (−21,053) and `Condition2_PosN` (−9,517). Strong positive signals also include quality and neighborhood indicators such as `GarageQual_Ex`, `YearBuilt`, `OverallQual`, `Neighborhood_NoRidge`, and `Neighborhood_StoneBr`. This aligns with domain expectations: size, build/finish quality, and location remain the strongest determinants of sale price in the selected sparse model.

Table 4: Top 10 feature effects loaded from `t20_feature_interpretation.csv`.

Feature	**Coefficient**	**Direction**	**Interpretation**
`GrLivArea`	29348.34	positive	Size-related feature: larger usable space generally increases price.
`RoofMatl_ClyTile`	−21052.55	negative	General structural/categorical feature that appears to decreases price.
`Condition2_PosN`	−9516.55	negative	Quality/condition proxy: better values typically decreases price.
`GarageQual_Ex`	9370.23	positive	Quality/condition proxy: better values typically increases price.
`YearBuilt`	8778.70	positive	Age/renovation signal: newer or renovated homes usually increases price.
`BsmtQual_Ex`	8553.76	positive	Quality/condition proxy: better values typically increases price.
`OverallQual`	8356.41	positive	Quality/condition proxy: better values typically increases price.
`PoolArea`	7508.56	positive	Size-related feature: larger usable space generally increases price.
`GarageCond_Ex`	−7325.71	negative	Quality/condition proxy: better values typically decreases price.
`BsmtFinSF1`	6571.54	positive	Size-related feature: larger usable space generally increases price.

6 Limitations

1. **No global convergence theorem for the outer loop.** The ISTA/FISTA convergence rates [Beck and Teboulle, 2009] apply to the inner fixed-weight subproblems. The outer IRL1 loop changes the objective each iteration, so standard convergence bounds do not apply globally. Outer-loop convergence is demonstrated

empirically, not proven theoretically.

2. **No oracle property claim for the dynamic variant.** The static Adaptive LASSO has oracle properties under conditions specified by Zou [2006] and extended by Huang et al. [2008]. The dynamic IRL1 variant does not inherit these guarantees automatically; extending oracle property analysis to the iterative reweighting case requires additional theoretical work.

3. **Efficiency claims are solver-relative.** Comparisons of runtime and iteration counts are meaningful only between custom ISTA/FISTA variants under matched stopping criteria. No claim is made that the custom implementation outperforms scikit-learn's production coordinate descent.

4. **Sensitivity to preprocessing.** Results depend on the choice of imputation strategy, encoding, and scaling. Different preprocessing choices may alter which features are selected and how stable that selection is.

5. **Single dataset.** All empirical conclusions are specific to the House Prices dataset. Generalisation to other high-dimensional tabular regression tasks requires further evaluation, particularly in the $p \gg n$ regime studied by Huang et al. [2008].

7 Conclusion

This project implements and empirically evaluates IRL1-PG—the Iteratively Reweighted ℓ_1 Minimisation framework of Candès et al. [2008] integrated within ISTA and FISTA proximal gradient solvers [Beck and Teboulle, 2009]—for sparse feature selection in high-dimensional tabular regression. The method is applied to the House Prices dataset, which exhibits the strong feature correlations that motivate dynamic reweighting over static ℓ_1 penalties [Zou and Hastie, 2005]. The IRL1 outer loop is understood within the broader class of reweighted sparsity methods unified by Wipf and Nagarajan [2011], while the FISTA acceleration follows Nesterov's [1983] momentum framework.

The key findings are: (1) dynamic IRL1-PG produces substantially sparser models (92–99 non-zeros) than Ridge, LASSO, and static Adaptive LASSO (240–283 non-zeros) while improving test error on this dataset split; (2) within dynamic variants, ISTA achieves the best validation MSE while FISTA achieves the best test MSE and MAE with comparable runtime; (3) dynamic reweighting improves sparsity–error trade-offs but reduces support stability across random seeds relative to static adaptive weighting. Limitations include the absence of a global outer-loop convergence proof and sensitivity to preprocessing choices.

Future directions include extending IRL1-PG to generalised linear models, combining it with group-sparse penalties for handling structured correlations [Zou and Hastie, 2005], and applying it to higher-dimensional genomics or financial datasets where the $p \gg n$ regime is more extreme [Huang et al., 2008].

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